Mohaimin Al Aoun

Senior Full-Stack Software Engineer
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Education

Vanderbilt University
Bachelor of Science, Computer Science 2017
Undergraduate Teaching Assistant, College of Engineering 2015 - 2017

Experience

Full-Stack Software Engineer II — **Nov 2021 - Present**
Qualtrics — **Reston, Virginia**

- **Architected Interaction Viewer**, a multi-dataset, multi-host platform product built collaboratively with PM and UX to replace a legacy Clarabridge tool, using Module Federation 2, React, and TypeScript with three federated entry points consumed by multiple host apps, with shared singleton dependency strategy to prevent version conflicts across teams.
- **Built an AI-powered dashboard migration service** using Claude API with forced structured output to generate per-widget migration plans, cutting Studio-to-Engage rebuild time from ~30 to ~5 minutes; built the full pipeline across Node.js, Express, SSE streaming, and a Python FastAPI proxy with HMAC JWT auth against the corporate LLM gateway. Selected as 1 of 5 award winners from 160 projects at Qualtrics Innovation Month, judged by the executive panel.
- **Built Node.js/Express Backend-for-Frontend (BFF)** orchestrating 5+ upstream platform APIs with parallel requests, JWT auth, intelligent retry logic, bounded timeouts, and graceful degradation, covered at 98% (Vitest + Supertest). Deployed as a multi-datacenter Kubernetes service with KEDA autoscaling, Prometheus metrics, and Vault secret management.
- **Optimized rendering performance** with virtual scrolling (TanStack Virtual) for 1,000+ record datasets; reduced redundant network calls via aggressive TanStack Query caching and batch API loading.
- **Published reusable NPM packages, CLI tools, and design system components** adopted by 10+ teams, cutting UI dev time ~30%; led post-acquisition design system unification reducing UI inconsistencies across 6 enterprise products.

Software Engineer — **March 2020 - Oct 2021**
Clarabridge (acquired by Qualtrics) — **Reston, Virginia**

- **Built internal React + Express tooling for NLP and audio analysis,** including a sentence-level POS/topic visualizer and audio timeline UI used by transcription, research, and product teams
- Created and maintained a shared **UI component library**, enabling seamless integration of accessible front-end components across multiple enterprise products and cutting dev time for new features by ~25%.
- **Modernized legacy codebase**, migrating core features from **AngularJS to Angula**r in a large-scale hybrid enterprise application, reducing tech debt and improving performance and maintainability.

Software Engineer — **June 2017 - March 2020**
MicroStrategy — **Tysons, Virginia**

- **Built core data visualization features** using React, Redux, and TypeScript for an enterprise BI platform serving thousands of analysts; launched Open Canvas enabling free-form overlay of complex visualizations
- **Developed proprietary JavaScript rendering framework** reducing dashboard load times by 40%; wrote automated tests with Protractor, Cucumber, and Chai improving regression coverage

Software Engineering Intern — **May 2016 - August 2016**
Digital Reasoning (Acquired by Smarsh) — **Franklin, Tennessee**

- **Built full-stack image annotation platform** (Node.js, AngularJS, REST API in Express) used to generate ML training data for object detection models

Personal Projects

Novagrapple: BJJ Instructor Platform deployed on Vercel and Railway

- **Developed and deployed a React (Rsbuild) instructor website builder** with real-time split-screen preview, 5 distinct visual themes, and live profile rendering at novagrapple.com/@username
- **Built a Spring Boot + MongoDB class scheduling platform** with instructor search, booking management, and an instructor dashboard
- **Integrated Firebase Auth and LLM-powered bio generation (OpenAI)** to support a two-sided marketplace connecting instructors with students.